

Name: Key

6. a) Give a basis for the column space of  $A$ ; or explain why one does not exist.

$$A = \begin{pmatrix} 1 & 2 & 10 & 5 \\ 2 & 2 & 8 & 6 \\ 3 & 1 & 0 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 10 & 5 \\ 2 & 2 & 8 & 6 \\ 3 & 1 & 0 & 5 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 10 & 5 \\ 0 & -2 & -12 & -4 \\ 3 & 1 & 0 & 5 \end{pmatrix} \xrightarrow{R_3 - 3R_1} \begin{pmatrix} 1 & 2 & 10 & 5 \\ 0 & -2 & -12 & -4 \\ 0 & -5 & -30 & -10 \end{pmatrix}$$

$$\xrightarrow{-\frac{1}{2}R_2} \begin{pmatrix} 1 & 2 & 10 & 5 \\ 0 & 1 & 6 & 2 \\ 0 & -5 & -30 & -10 \end{pmatrix} \xrightarrow{R_3 + 5R_2} \begin{pmatrix} \boxed{1} & 2 & 10 & 5 \\ 0 & \boxed{1} & 6 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \text{ REF.}$$

Since the first and ~~third~~<sup>second</sup> columns of this matrix are a basis for its column space, the same is true for the first and second columns of  $A$ .

So a basis for  $\text{col}(A)$  is  $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \right\}$

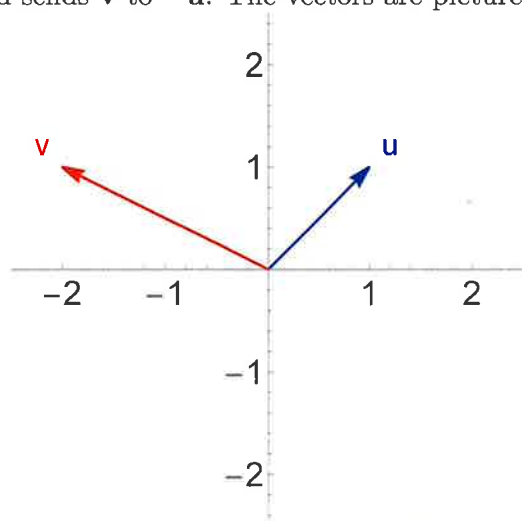
b) Give an example of a basis for  $\mathbb{C}^3$  that contains the two vectors  $\begin{pmatrix} 0 \\ 1 \\ i \end{pmatrix}$  and  $\begin{pmatrix} 0 \\ 3i \\ -3 \end{pmatrix}$ , or explain why this is not possible.

The vectors in a basis are linearly independent.

But  $\begin{pmatrix} 0 \\ 3i \\ -3 \end{pmatrix} = 3i \begin{pmatrix} 0 \\ 1 \\ i \end{pmatrix}$ . So these two vectors cannot be in a basis for  $\mathbb{C}^3$  together.

Name: Key

7. a) Using the standard basis, give the matrix for the linear transformation of  $\mathbb{R}^2$  that sends  $u$  to  $v$  and sends  $v$  to  $-u$ . The vectors are pictured below, in the standard coordinate system.



In the basis  $u, v$  this linear transformation is represented by the matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

To write this linear transformation

in the standard basis, apply a change of basis:

$$\begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1/3 & 2/3 \\ -1/3 & 1/3 \end{pmatrix}$$

convert back  
from  $u, v$  basis  
to standard basis

convert from  
standard basis  
to  $u, v$  basis

$$= \begin{pmatrix} -2 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1/3 & 2/3 \\ -1/3 & 1/3 \end{pmatrix}$$

$$= \begin{pmatrix} -1/3 & -5/3 \\ 2/3 & 1/3 \end{pmatrix}$$

b) Recall that a diagonal matrix is a square matrix such that all of its non-zero entries are on its main diagonal (upper left to bottom right). Give an example of a matrix that is similar to a diagonal matrix but that is not itself diagonal; or explain why this is not possible.

For instance,  $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$  is diagonal but  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is not.

Both are matrices representing a reflection in a line in  $\mathbb{R}^2$  and so are similar. There are many other examples; doing a change of basis to a diagonal matrix does not in general keep it diagonal:  ~~$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$~~   $\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}^{-1}$

Check of part (a):

$$\begin{pmatrix} -1/3 & -5/3 \\ 2/3 & 1/3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \stackrel{=u}{=} \begin{pmatrix} -2 \\ 1 \end{pmatrix} \checkmark = v$$

$$\begin{pmatrix} -1/3 & -5/3 \\ 2/3 & 1/3 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} \stackrel{=v}{=} \begin{pmatrix} -1 \\ -1 \end{pmatrix} \checkmark = -u$$

$\begin{pmatrix} 7 & -3 \\ 10 & -4 \end{pmatrix}$   
 diagonal  $\uparrow$  not diagonal